ID	Non-Functional Requirement	Priority	Category	Acceptance Criteria
NFR-01	The system shall return blood-availability search results within 2 seconds under normal operating	High	Performance	At least the defined normal test workload returns results within 2 seconds.
NFR-02	The system shall require authentication for protected functions and shall allow only authorized blood-bank staff to	High	Security	Unauthenticated/unauthorized users cannot access protected inventory functions.
NFR-03	The system shall protect the personal information of donors and users from unauthorized access.	High	Privacy	Unauthorized users cannot access protected personal information.
NFR-04	The system shall maintain accurate blood-inventory information and shall prevent the available blood quantity	High	Reliability	Inventory never falls below zero after valid or invalid update attempts.
NFR-05	The system shall be available during the defined operating hours of registered blood banks except during scheduled	High	Availability	Users can access the system during defined operating hours except during documented maintenance.
NFR-06	The system shall provide a simple and user-friendly interface that allows users to search for blood availability without	Medium	Usability	A normal user can complete a blood-availability search without technical assistance.
NFR-07	The system shall support multiple blood banks, donors, donations, and inventory records without requiring	Medium	Scalability	Additional records/blood banks can be added without architectural modification.
NFR-08	The system shall use a modular structure so that individual components can be maintained or modified without	Medium	Maintainability	Individual modules can be modified and tested independently where applicable.
NFR-09	The system shall perform regular backups of important donor, donation, and inventory data and shall provide a	High	Backup & Recovery	Backup can be created and stored, and data can be restored using the recovery mechanism.
NFR-10	The system shall operate correctly on commonly used modern web browsers such as Chrome, Edge, and Firefox.	Medium	Compatibility	Core system functions operate correctly on the specified browsers.
NFR-11	The system shall display clear error messages for invalid inputs or unsuccessful operations and shall not	High	Error Handling	Invalid input produces an appropriate error message and does not terminate the application.
NFR-12	The system shall record important activities such as inventory updates, donation records, and administrative	High	Auditability	Each specified activity creates an audit record containing user, action and timestamp.
NFR-13	The system shall maintain consistent and accurate donor, donation, and inventory records and shall prevent	High	Data Integrity	Duplicate IDs and invalid inventory values are rejected and records remain consistent.